Luis Garcia Rivera

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Education

Washington University in St. Louis, McKelvey School of Engineering — St. Louis, MO
B.S. Mechanical Engineering; Minor in Robotics & Mechatronics — Expected Dec 2027
GPA: 3.79/4.00 | Dean's List | HSF Scholar | Kessler Scholar

Engineering Experience

Manufacturing Equipment Engineer Co-Op – Tesla — **Fremont, CA**
Model 3 Body in White Manufacturing — *Jan 2026 – Aug 2026*

- Redesigned bowl-feeder EOAT locating hardware after a feeder installation blocked a critical locator; created 2 variants for 3 fixtures using NAAMS and custom 17-4 PH components to restore full 6-DOF part constraint; carried the design through machining, finishing, assembly, and installation, then passed a **25-part live-line validation with no issues**; tooling is now running in production.
- Diagnosed a clamp-misalignment fault on the same EOAT/bowl-feeder line that let parts slide past the second locating pin; redesigned the clamp with a weld-bead clearance chamfer, machined it in-house, and installed it, substantially reducing the fault in production.
- Partnered with two senior engineers to cut station cycle time from **75s to 65s (13%)** in a three-robot FANUC spot-weld sequence; reprogrammed an adjacent robot in teach mode to assume one weld and its approach/home points from the bottleneck robot, then validated interference, full-speed operation, and production stability with maintenance.
- Reverse-engineered an OEM Daifuku three-clamp fixture system used at 8 production locations into internal CATIA V5 models and GD&T drawings, creating an in-house fabrication path versus an approximately **$88K replacement package** and quoted 10-month lead time.

Optimus Training Data Collection – Modular Factory Hardware

- Co-developed and deployed modular training-data hardware to **20 mounted Fremont stations**, targeting a three-person-to-one-operator workflow; co-led surveys, assembly, wiring, commissioning, troubleshooting, and camera-angle iteration around live production.
- Owned CATIA V5 mechanical design for **3 adjustable camera-mount architectures** and tool-free, 3D-printed compute-module and battery-pack mounts with locating pins and swappable interfaces for 80/20 extrusion, I-beams, and drilled surfaces; released designs to a print farm and standardized hardware for repeatable deployment and handoff.

Optimus Academy – Humanoid Robotics Training Facility

- Owned the layout, equipment-tracking, and demolition-documentation workstream for a planned 40-station training facility; produced **21 contractor-ready station layouts** with 179 equipment callouts and coordinated execution with architects, facilities, controls engineers, and contractors; integrated conveyor/VFD, conduit, and compressed-air requirements into the permitted layout.

Undergraduate Researcher – Computational Mechanics & AI Lab — **St. Louis, MO**
Applied Machine Learning in Fluid Dynamics — *May 2025 – Aug 2025*

- Developed physics-informed neural networks in PyTorch for cardiovascular flow prediction, achieving up to a $1000\times$ **reduction in prediction time**; modernized a legacy Fortran solver into a modular Python pipeline for reproducible validation and rapid parameter studies.

Structures Engineer – WashU Design/Build/Fly (AIAA) — **St. Louis, MO**
Composite Airframe Design and Test — *Aug 2024 – Present*

- Designed load-bearing composite airframe structures in SolidWorks and used ANSYS finite-element analysis to validate safety factors while minimizing takeoff weight; led fabrication and destructive testing to correlate physical failures with simulation predictions.

Technical Skills

Mechanical Design & Manufacturing: CATIA V5/3DEXPERIENCE, SolidWorks (CSWA), AutoCAD/DWG, GD&T, DFM/DFA, fixtures and end-of-arm tooling, 3D printing, mill/lathe machining, composite fabrication
Automation & Test: FANUC/KUKA robot programming and teach-pendant operation, Allen-Bradley GuardLogix PLC, SCADA, pneumatics, sensor integration (SICK/Keyence, proximity, laser, light curtains)
Analysis & Software: ANSYS FEA/CFD, Python (PyTorch, NumPy, Pandas, Matplotlib), C++/Arduino, MATLAB, root-cause analysis, DOE, OEE optimization

Project

Autonomous Mobile Robot – Embedded Controls Project — Sep 2024 – Dec 2024

- Designed and prototyped an Arduino/C++ autonomous vehicle integrating sensors, a DC-motor drivetrain, and PID control; achieved **95% path-following accuracy** in autonomous operation.

Leadership & Activities

Teaching Fellow: Engineering Circuits | **Tutor:** WashU Learning Center | **Undergraduate Advisory Board**